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Review

Nutritional rickets around the world☆

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ABSTRACT

Nutritional rickets is a major public health problem in many countries of the world. The disease is characterized by deformities of the long bones, enlargement of the wrists and costochondral junctions, hypotonia and, in infants, craniotabes and delayed fontanelle closure. Predominantly caused by severe vitamin D deficiency, rickets can also be associated with hypocalcemic seizures and cardiac failure. First presentation is typically at 6–24 months of age, although hypocalcemia may be evident in younger infants. In many affluent industrialized countries, the prevalence of rickets in the general population diminished after the introduction of clean-air legislation and dietary supplementation. However, in such countries, vitamin-D deficiency rickets has re-emerged in recent years, particularly among groups with limited exposure to UVB-containing sunshine. Infants at risk of rickets tend to be those whose mothers had poor vitamin D status during pregnancy and those exclusively breast-fed for a prolonged period with little skin exposure to UVB. In other countries of the world, the prevalence of rickets can be high, even in regions with abundant year-round UVB-containing sunshine. In general, this is also due to vitamin D deficiency related to limited sun exposure. However, reports from Africa and Asia suggest that there may be other etiological factors involved. Studies in South Africa, Nigeria, The Gambia and Bangladesh have identified rickets in children, typically 3–5 years old at first presentation, in whom plasma 25-hydroxyvitamin D concentrations are higher than those characteristic of primary vitamin D deficiency. Calcium deficiency has been implicated, and in some, but not all, disturbances of phosphate metabolism, renal compromise and iron deficiency may also be involved. Continuing studies of the etiology of nutritional rickets will provide evidence to underpin guidelines for the prevention and treatment of rickets world-wide.

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1. Rickets

Rickets is a disease of children characterized by a failure or delay in endochondral calcification of the growth plates of long bones [1]. This results in widening and splaying of the growth plates and leads to enlargement of the wrists and costochondral junctions, and the characteristic deformities of the lower limbs, notably genu varum (bow legs) and genu valgum (knock-knees). Rickets is generally

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Table 1
Classification of rickets by biochemical profile.

Type	Examples	Biochemical profile
Calciopenic	Vitamin D deficiency Dietary calcium deficiency	PCa ↓ PTH ↑ Urinary Ca ↓
Phosphopenic	Urinary phosphate loss with FGF23 abnormalities: Genetic disorders e.g. XLH, ADHR Tumor-associated rickets Urinary phosphate loss without FGF23 abnormalities: Fanconi syndrome Distal renal tubular acidosis Dietary phosphorus deficiency	PPO_4 ↓ PCa ↔ PTH ↔ FGF23 ↑ or ↔ Urinary TmP/GFR: Renal causes: ↓ Dietary phosphorus deficiency/intestinal causes: ↑
Inhibited mineralization	Hereditary hypophosphatasia Aluminum, fluoride toxicities Bisphosphonates (1st generation)	PCa, PPO_4

Table modified from [2], not all the causes of rickets have been included.
PCa: plasma calcium, PPO_4: plasma phosphate, PTH: parathyroid hormone, FGF23: fibroblast factor 23, TmP/GFR: tubular maximum reabsorption of phosphate, XLH: X-linked hypophosphatemic rickets, ADHR: autosomal dominant hypophosphatemic rickets.

accompanied by osteomalacia, i.e. defective mineralization of preformed osteoid in bone, fragility fractures and muscle weakness [2]. Young infants present with craniotabes and delayed fontanelle closure, hypotonia, hypocalcemia, seizures and cardiac failure [1].

Rickets can be classified into three main types depending on the biochemical profile: calciopenic, phosphopenic and inhibited mineralization [2,3], as described in Table 1. Typically, calciopenic rickets is associated with an elevated plasma parathyroid hormone concentration (PTH) in response to low plasma calcium, which results in internalization of phosphate transporters in the kidneys and decreased renal phosphate reabsorption. Phosphopenic rickets is associated with a chronically low plasma phosphate with normal PTH and results from increased production or gain-of-function of FGF23, a phosphaturic hormone, or from renal disorders that compromise phosphate reabsorption. In both cases the result is urinary phosphate loss and hypophosphatemia, leading to reduced apoptosis of hypertrophic chondrocytes in the growth plate and rickets. The third category, inhibition of mineralization, refers to rickets where the mineralization process in the growth plate is directly affected and typically the plasma concentrations of calcium and phosphate are normal. There are a variety of underlying causes responsible for the various forms of rickets, including genetic disorders, such as autosomal dominant hypophosphatemic rickets (ADHR) and X-linked hypophosphatemic rickets (XLH), tumors, organ malfunction, drugs and exposure to toxic agents such as aluminum and fluoride. This review will focus on rickets of public health concern caused by nutritional deficiencies, referred to as 'nutritional rickets'.

2. Rickets prevalence world-wide

Rickets was common in Europe until the mid- 20th century and was known as "The English Disease" because of its high prevalence in England. The seminal work of pioneers such as Harriette Chick and Elsie Dalyell demonstrated that this was due to vitamin D deficiency [4]. In their studies in a children's hospital in Vienna immediately after World War I, they demonstrated that rickets could be prevented and cured by being in the sunlight outdoors, in the summer not winter, or by exposure to mercury vapour quartz lamp and by cod liver oil, which is now know to contain cholecalciferol, vitamin D_3. Studies by Hume and Steenbock in rats further demonstrated the antirachitic activities of UVB radiation and some irradiated foods, now known to contain ergocalciferol, vitamin D_2 [4].

Since the 1950s, rickets has become far less common in the UK as a result of clean air legislation, food fortification and public awareness [5]. The National Diet and Nutrition Surveys and other data from the UK, however, have shown that sub-groups of the population in the UK continue to be at increased risk of vitamin D deficiency, particularly among its substantial South Asian ethnic minority [5–7]. These people, or their forebears, originally migrated from India, Pakistan and Bangladesh and often retain the close family networks and traditional diet and clothing customs of South Asia. In recent years, a resurgence of rickets cases has been reported from the major cities in the UK and other Northern European countries particularly among children from families of South Asian, African, Afro-Caribbean or Middle-Eastern origin [8–10]. Commonly, affected children are reported to be dark-skinned, to have had limited opportunities for skin UVB exposure, were born to mothers who were likely to have had poor vitamin D status during pregnancy, were exclusively breast-fed for an extended period and weaned onto poor diets [3].

There are currently no national statistics on the prevalence of nutritional rickets in the UK but there is considerable concern that this is a widespread problem despite public health recommendations for pregnant women, children and those with restricted sunshine exposure to consume a daily vitamin D supplement [5]. In Denmark, another northern European country with a sizeable immigrant population, a population-based retrospective estimate of the burden of nutritional rickets from medical records has confirmed a higher incidence of nutritional rickets among ethnic minorities, most commonly children of parents originating in Africa and the Middle East [11]. The recorded annual incidence in 1995–2005 among ethnic Danish children aged 0–2.9 years was 2.0 per 100,000 per year (less than the incidence of hereditary rickets of 4.3 per 100,000 per year) while that among those from immigrant families born in Denmark was fifty times greater at 100 per 100,000 per year.

Understandably, much of recent literature on the prevalence of nutritional rickets has come from those industrialized countries experiencing a resurgence of clinic presentations with rickets. However, the greatest burden of disease world-wide appears to be in Africa, the Middle East and Asia [2,6,12], where prevalence rates of 10–70% have been reported, as summarized in Table 2. It is difficult to compare such reported prevalence rates directly because of the different methodologies employed. Many of the estimates are based either on population surveys that recorded the presence of bone deformities and other signs consistent with rickets or on hospital records of rickets among patients admitted to hospital with acute diseases such as pneumonia (Table 2 [2,13–19]). This contrasts with clinic-based records of radiographically-confirmed rickets in countries where children in the general population have ready access to routine health-care. Some more recent estimates from Asia and Africa that have combined population surveys with confirmation by physician examination or X-rays have reported somewhat lower prevalence rates [20–22]. However, even the most conservative estimates provide evidence that prevalence rates in these countries are greater than in Western countries and are of a magnitude that represents a global public health concern.

3. Causes of nutritional rickets world-wide

Vitamin D deficiency appears to the major factor underlying nutritional rickets in many countries world-wide [23,24], even

Table 2
Examples of reported prevalence of rickets in Africa, Middle East and Asia.

Country	Year	Rate%	Sample	Method
Mongolia [13]	1998	70	*P* < 5 years	Rickets signs
Tibet [14]	1994	66	*P* < 2 years	Rickets signs
Ethiopia [15]	1997	42	*H** < 5 years	X-rays
Yemen [16]	1987	27	*V* < 5 years	–
Turkey [17]	1994	10	*P* < 3 years	–
Nigeria [18]	1998	9	*P* < 3 years	Rickets signs
Iran [19]	1975	15	*H* < 5 years	X-rays
China [20]	1977–83	47	*P* < 3 years	Rickets signs
		3.7		X-rays/biochem
The Gambia (West Kiang) [21]	2007	3.3	*P* < 18 years	Rickets signs
		0.6		Physician exam
Bangladesh (Chittagong) [22]	2008	2.2	*P* < 16 years	Rickets signs
		1.0		X-rays

P: population sample, *H*: hospital-based sample, *H**: children admitted to hospital with pneumonia, V: children attending for vaccination.

in 'sun-rich' countries where there is year-round opportunity for skin exposure to UVB-containing sunlight [25–27]. Vitamin-D deficiency rickets is often first diagnosed in infancy or young childhood and is frequently associated with limited skin sunshine exposure of the mother and child [1]. There are many reasons for such limitations, including wearing clothing that covers most of the body for cultural, religious or climatic reasons, and living predominantly indoors. Such children generally display the classical manifestations of vitamin D deficiency rickets relevant to their age, which can include craniotabes, open fontanelle, rachitic rosary and Harrison's sulcus [1,3]. They also have a calciopenic rickets biochemical profile of severe hypovitaminosis D (plasma 25-hydroxyvitamin D concentration (25OHD) <25 nmol/l (10 ng/ml) but usually <12.5 nmol/l (5 ng/ml)), hypocalcemia and elevated PTH [1,3].

However, it is clear from a number of reports that primary vitamin D deficiency does not adequately describe the nutritional rickets seen in a number of African, Middle Eastern and Asian countries [2,18,26,28–31]. In such cases, rickets is not associated with severe hypovitaminosis D and, although the levels may be lower than those of their peers in the local community, the children generally have 25OHD concentrations that are similar to those recorded in surveys of Western children. The age at diagnosis is older, generally around 3–5 years, and some of the classical signs of vitamin D deficiency, such as rachitic rosary and Harrison's sulcus, are often absent. It is likely, therefore, that factors other than vitamin D deficiency are involved in the etiology of nutritional rickets in these countries.

4. Calcium-deficiency rickets

Reports from South Africa and Nigeria have suggested that calcium deficiency is a major factor in the etiology of rickets in those countries [2,29], and have demonstrated that some affected children respond well to treatment with calcium alone [32].

A few years ago we reported a case series of patients in The Gambia with bone deformities consistent with rickets who had a similar clinical profile at presentation to that reported from Nigeria and South Africa [28]. The Gambia is a tropical West African country (latitude 13°N) where most children have a very low calcium intake and abundant, unrestricted UVB-sunlight exposure. The children summarized in the case series had a mean age of 3.4 years (range 1.1–16.4 years), 65% were boys and 28% had active rickets as defined radiologically using the Thacher scoring system [33], an elevated plasma alkaline phosphatase activity or both. The typical biochemical profile at presentation was a 25OHD above 25 nmol/l (10 ng/ml), but lower than other children in the community; an elevated plasma 1,25-dihydroxyvitamin D concentration (1,25$(OH)_2$D) compared to local children, who themselves had higher 1,25$(OH)_2$D concentrations than those seen in Western children of similar age; and low plasma phosphate and elevated PTH in those children with active rickets (Fig. 1). There was no difference in plasma calcium concentration. No urine samples were available. The majority of these children, whether classified as having active or non-active rickets, had a very high plasma concentration of FGF23 which was inversely correlated with plasma phosphate concentration. This suggested that a disturbance of phosphate metabolism may have contributed to their rickets etiology (Fig. 1). Some of the FGF23 concentrations observed were of a magnitude only previously reported in children with genetic disorders, such as X-linked hypophosphatemia, or gross pathology, such as tumor-induced osteomalacia or chronic kidney disease, and remained elevated over a 12 month period of treatment and follow-up [28].

The patients in this case series were prescribed daily high-dose vitamin D supplementation by the local physicians at a dose that depended on age. This averaged 10,000 IU/d for 6 months and was coupled with a calcium lactate supplement that provided an extra 117 mg calcium/d [28]. The response to this treatment, illustrated in Fig. 2, did not correspond to that typical of a child with primary vitamin D deficiency [2]. Most notably, at the end of the 6 month period of supplementation 1,25$(OH)_2$D had not decreased to normal concentrations but rather was approximately 3-fold higher than local children and remained higher than at presentation when measured 6 months after the vitamin D therapy finished. The concentration of 25OHD was also increased substantially at 6 months in all patients but by 12 months had returned to the lower concentration relative to local children seen at presentation. By 6 and 12 months, the hypophosphatemia had resolved and, although FGF23 remained elevated for many of the children, there were no significant correlations evident between plasma phosphate and FGF23 at either timepoint. There was no evidence of hypercalcemia in the treated children.

These data led to the hypothesis that the rickets suffered by these children was due to chronic calcium deficiency, which had resulted in elevated PTH and 1,25$(OH)_2$D, stimulation of FGF23 production, decreased renal phosphate reabsorption, increased urinary phosphate excretion and lowered plasma phosphate [28]. In such a scenario, the low plasma phosphate would further stimulate the renal 1-alpha-hydroxylase enzyme CYP27B1 to produce 1,25$(OH)_2$D. In a healthy child with an adequate diet, such a perturbation would lead to increase in calcium and phosphate absorption through the increased activity of 1,25$(OH)_2$D on the intestine, homeostasis would be regained and PTH, 1,25$(OH)_2$D and FGF23 concentrations would return to normal [34]. However, in the absence of a sufficient calcium supply, the continued stimulation of PTH by the lack of calcium plus the lowered plasma phosphate due to the elevated FGF23 [35] would mutually reinforce the signals on CYP27B1 to increase 1,25$(OH)_2$D synthesis, sufficient to override the suppressive effects of FGF23 on this 1-alpha-hydroxylase. This

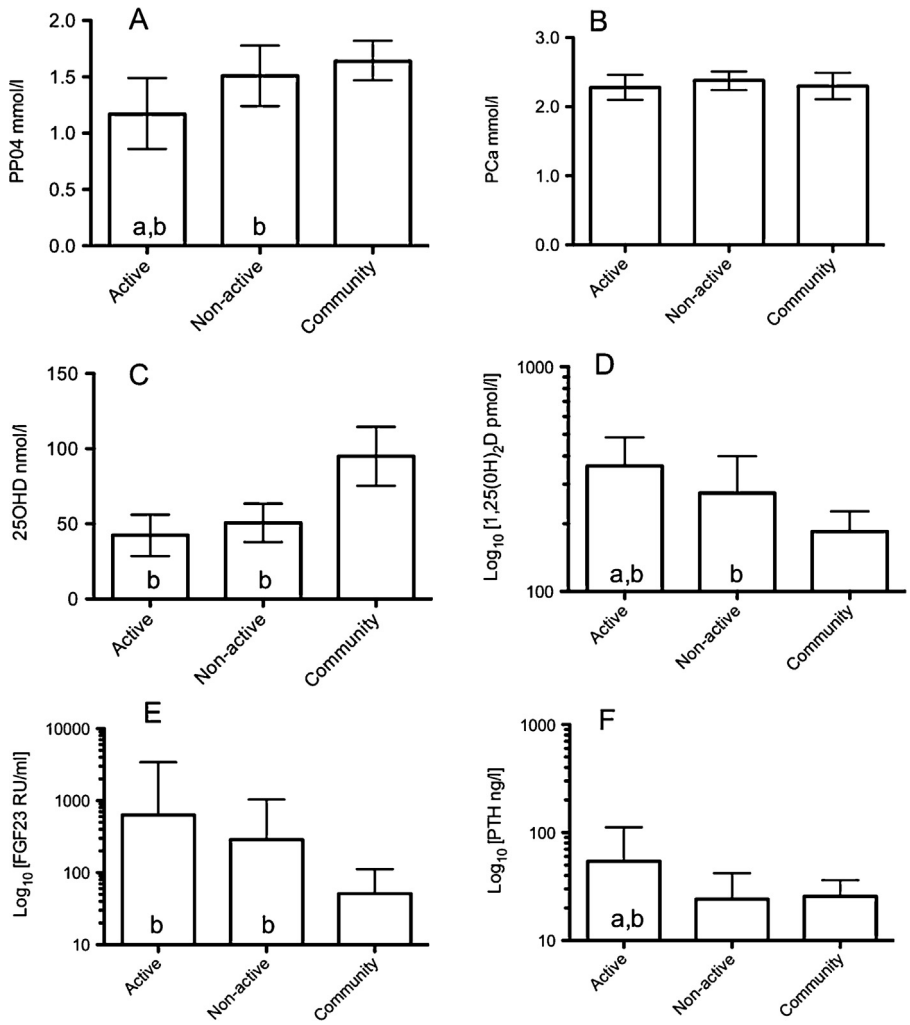

Fig. 1. Plasma biochemistry of Gambian rickets patients at presentation and community reference children.
Data are mean ± SD from [28]. Active rickets, $n = 13$; non-active rickets, $n = 33$; Community reference children, $n = 147$ except for FGF23, $n = 30$. (A) phosphate in mmol/l; (B) calcium in mmol/l; (C) 25-hydroxyvitamin D in nmol/l; (D) 1,25-dihydroxyvitamin D in pmol/l; (E) fibroblast factor-23 in RU/ml; (F) parathyroid hormone in ng/l. Graphs (D), (E) and (F) are on a logarithmic scale. Comparison of pairs of subject category evaluated by Scheffé post hoc tests using Linear Model software, Datadesk 6.3: (a) active rickets group significantly different from non-active rickets group, $p \leq 0.01$, (b) active rickets group or non-active rickets group significantly different from community reference children $p \leq 0.01$.

would result in continued FGF23 production and urinary phosphate loss, ultimately leading to rickets [28]. Such a mechanism could also explain the lower 25OHD concentrations seen in these children, because, not only would there be greater conversion of 25OHD to $1,25(OH)_2D$, but also stimulation of the renal 24-hydroxylase enzyme CYP24A1 by FGF23 [36] would increase conversion of 25OHD to its more abundant metabolite, 24,25-dihydroxyvitamin D ($24,25(OH)_2D$) and of $1,25(OH)_2D$ to $1,24,25(OH)_3D$.

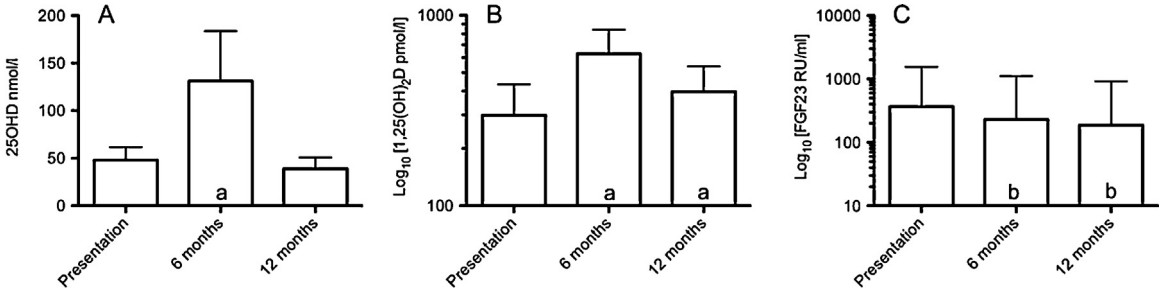

Fig. 2. Plasma biochemistry of Gambian rickets patients at presentation and follow-up.
Data are mean ± SD from [28].
(A) 25-hydroxyvitamin D in nmol/l; (B) 1,25-dihydroxyvitamin D in pmol/l; (C) fibroblast factor-23 in RU/ml. Graphs (B) and (C) are on a logarithmic scale. Comparison of follow-up measurement with presentation value within individuals evaluated by Scheffé post hoc tests using Linear Model software, Datadesk 6.3: (a) significantly different from presentation, $p \leq 0.001$, and (b) significantly different from presentation, $p \leq 0.05$.

5. Nutritional rickets of mixed etiology

The hypothesis we proposed to explain the etiology of rickets in The Gambia neatly described the presenting biochemical features and response to vitamin D treatment. However, it could not explain why some Gambian children develop rickets while the majority do not, given that most children in that country have very low calcium intakes, of around 200–300 mg/d [37]. Similarly in Nigeria, low calcium intakes alone were not considered to fully account for the rickets seen in that country [29]. In children with a low calcium intake, marginal vitamin D status, i.e. 25OHD concentrations above those traditionally associated with vitamin-D deficiency rickets, might provide an additional precipitating factor, given the potentially greater vitamin D supply that might be required to support the increased 25OHD conversion to its metabolites, as described above. This possibility was suggested by a cohort of rickets patients in the US, where a spectrum of low calcium intake and poor vitamin D supply together or separately described the symptomatology [38] and by the Nigerian reports [39]. However, restricted vitamin D supply is an unlikely discriminating factor in The Gambia where there is year-round UVB in sunlight, the climate for much of the year is hot and dry with relatively clear skies, all children participate in outdoor pursuits on a daily basis, and clothing for young children does not limit skin exposure to sunlight on the face, arms, lower legs and often other parts of the body [6].

To consider the possibility of predisposing factors, we invited the Gambian children who had been described in the original case series to participate in an intensive follow-up study approximately 5 years after their original presentation at clinic and compared them to local children of the same age and sex with no history of rickets [40]. At follow-up, 54% had lower limb deformity and 19% had raised FGF23. No family links between the patients were evident, except for one pair of siblings. The study demonstrated modest but significant differences between the children with a history of rickets and local children: their calcium intake and dietary calcium-to-phosphorus ratio were lower, related to a smaller intake of cow's milk, and they had a lower tubular maximum reabsorption of phosphate (TmP/GFR), greater urinary fasting clearance and non-fasting 24 h excretion of phosphorus, and lower urinary fasting clearance and 24 h excretion of calcium [40]. These findings supported the calcium deficiency hypothesis. However, intriguingly, the children with a history of rickets also had a lower glomerular filtration rate as estimated from plasma cystatin C, suggesting a possible interaction with kidney function, and exhibited an inverse relationship between FGF23 and hemoglobin, suggesting a link with iron deficiency which is endemic in The Gambia [40].

We considered the possibility of a link with iron status more fully by expanding the study to include the siblings of the rickets children and a greater number of local children [41]. This confirmed the inverse correlation between FGF23 and hemoglobin, which was evident both in children from affected families and those from the local community. However, the slope of this relationship was significantly steeper in the children with a personal or family history of rickets than the local children (test for interaction $p < 0.0001$). The children from the affected families also had a greater prevalence of anemia, suggesting they carried a degree of predisposition from genetic or environmental factors that placed them at greater risk of rickets and anemia. These findings support the growing evidence suggesting the involvement of iron in FGF23 metabolism [42]. In a separate study of Gambian children without rickets, FGF23 was shown to decrease after iron supplementation, a change that correlated with improvements in hemoglobin, ferritin and zinc protoporphyrin [43]. These results were strengthened rather than diminished by considering possible confounding effects by the acute phase response to infection/inflammation [44]. Although at a lower prevalence than the rickets children, this research also demonstrated that a number of Gambian children without a history of rickets had an elevated FGF23 concentration that was not correlated with plasma phosphate [41], raising questions about FGF23 biology in children living in such an environment that merit further research. One possibility is that these elevated FGF23 concentrations, measured using the C-terminal Immunotopics assay, are due to high levels of circulating fragments, although the presence of fragments Gambian plasma samples was not detected by western blotting analysis [45].

It appears increasingly likely that, while vitamin D deficiency is the primary cause of nutritional rickets in many parts of the world, multiple factors may underlie the etiology of this disease in some populations. The Nigerian and South African studies have demonstrated that calcium deficiency is an important determinant. The Gambian studies have suggested elevated FGF23, less efficient kidney function and iron deficiency as possible candidates in addition to very low calcium intakes, although elevated FGF23 has not been reported from other populations at risk of calcium deficiency [46]. Chronic calcium deficiency may be due to very low intakes, but may also be due to poor absorption as a result, for example, of competing dietary factors, enteropathy and intestinal bacterial or parasitic infestations. Genetic susceptibility cannot be ruled out, for example differences in the distribution of vitamin D receptor alleles have been reported between rickets patients and local children in Nigeria and Turkey [26,47]. It is also possible that depletion of nutrients in the soil through intensive agriculture or flooding, or exposure to the toxic effects of natural, agricultural or industrial environmental contaminants, such as arsenic, organophosphates and lead, may be involved in the etiology of rickets in some populations. Bangladesh has one of the highest prevalence rates of nutritional rickets in the world, especially the Chakaria region between Chittagong and Cox's Bazar [22]. Ongoing studies in Chakaria by my research group are building on those of others [31] in order identify whether some or all of these factors are responsible for the epidemic of rickets experienced by children in that population.

6. Concluding remarks

There is an unacceptably high world-wide prevalence of nutritional rickets, a preventable disabling disease of children. The greatest burden is in Africa, the Middle East and Asia, and, increasingly, in families originating from these countries resident in Europe, Australasia and the Americas. For many children, vitamin D deficiency is the predominant cause and is associated with limited UVB exposure by the mother before and during pregnancy and subsequently by the child. For such children the prevention and treatment of this condition are well defined [1] but poorly implemented or understood by health professionals or the population groups affected. However, primary vitamin D deficiency does not account for the nutritional rickets of many parts of Africa, the Middle East and South Asia, where calcium deficiency and other etiological factors are implicated. In such populations preventative strategies based on improving vitamin D status will not be effective and the children may not respond to vitamin D treatment. Continuing research into the etiology of nutritional rickets is needed to provide evidence to underpin guidelines for the prevention and treatment of rickets world-wide.

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